Hierarchical Uncertainty Quantification for Reliable Automated Radiology Report Generation

Cuantificación de Incertidumbre Jerárquica para la Generación Confiable de Informes Radiológicos Automatizados

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Abstract

With thousands of chest X-rays processed daily at facilities such as Khyber Teaching Hospital (KTH) and Lady Reading Hospital (LRH) in low-resource areas like Khyber Pakhtunkhwa, Pakistan, automated radiology report generation holds transformative potential for over-extended diagnostic services. Deep-learning vision-language models currently produce reports deterministically, without confidence scores, which can endanger patient safety through hallucinated clinical findings. This paper introduces a three-level hierarchical uncertainty quantification (UQ) framework that assigns calibrated confidence scores at word, sentence, and section granularity. The architecture combines a Vision Transformer (ViT-Base/16) visual encoder with a GPT-2 language decoder and stochastic inference via Monte Carlo (MC) Dropout. Trained and evaluated on three publicly available Kaggle Chest X-ray datasets (combined $n = 5{,}863$ images), the system achieves BLEU-4 of 0.378 and CIDEr of 0.672, and correctly identifies over 94.3% of erroneous clinical statements at a safety threshold of $\tau = 0.75$. The Area Under the Calibration Error curve (AUCE) is 0.042, indicating strong probabilistic calibration. Results confirm that hierarchical uncertainty estimation can be integrated into automated radiology pipelines without degrading linguistic performance, providing a viable pathway toward trustworthy AI-assisted diagnosis in under-resourced hospitals across Peshawar and beyond.

Keywords: automated radiology report generation, chest X-ray, deep learning, Monte Carlo Dropout, uncertainty quantification, Vision Transformer

Resumen

Con miles de radiografías de tórax procesadas diariamente en centros como el Hospital Universitario de Jaiber (KTH) y el Hospital Lady Reading (LRH) en zonas de bajos recursos como Jaiber Pajtunjwa, Pakistán, la generación automática de informes radiológicos tiene un potencial transformador para los servicios de diagnóstico sobrecargados. Los modelos de visión-lenguaje actuales producen informes de manera determinística, sin puntuaciones de confianza, lo que puede comprometer la seguridad del paciente mediante hallazgos clínicos alucinados. Este artículo introduce un marco de cuantificación de incertidumbre (UQ) jerárquico de tres niveles que asigna puntuaciones de confianza calibradas a nivel de palabra, oración y sección. La arquitectura combina un codificador visual Vision Transformer (ViT-Base/16) con un decodificador de lenguaje GPT-2 e inferencia estocástica mediante Monte Carlo Dropout. El sistema logra un BLEU-4 de 0,378 y un CIDEr de 0,672, e identifica correctamente más del 94,3 % de los enunciados clínicos erróneos con un umbral de seguridad de $\tau = 0{,}75$, con una curva AUCE de 0,042.

Palabras claves: aprendizaje profundo, cuantificación de incertidumbre, generación automática de informes radiológicos, Monte Carlo Dropout, radiografía de tórax, Vision Transformer

1 Introduction

Medical imaging diagnosis is one of the most cognitively demanding tasks in health care. Radiologists must analyze chest X-rays, CT scans, and MRI images, then produce structured reports that guide clinical decision-making. The worldwide deficit of radiologists has become a critical problem: the physician-to-population ratio falls as low as one radiologist per 50,000 inhabitants in many developing countries, including Pakistan [1].

Referral hospitals such as Khyber Teaching Hospital (KTH) and Lady Reading Hospital (LRH) in Peshawar each receive 1,200–1,800 chest X-rays per working day with fewer than eight

radiologists on staff, causing diagnostic queues that delay treatment and worsen outcomes [2].

Since the early 2010s, automated radiology report generation has been continuously investigated. Retrieval-based approaches matched query images to pre-written templates [3]. More recently, end-to-end encoder-decoder networks have demonstrated strong performance: Chen et al. introduced R2Gen, a memory-driven Transformer that set benchmark BLEU scores on IU-X-Ray and MIMIC-CXR [4]. Subsequent work incorporated medical knowledge graphs [5], cross-modal attention mechanisms [6], multi-scale visual features [7], and keyword-driven decoders [8]. However, a fundamental limitation persists across nearly all published systems: they are deterministic. A single point-estimate report is produced per image, with no measure of output reliability or trustworthiness.

This property is clinically hazardous. When a clinician relies on an automated summary, a model that hallucinate a finding—presenting an erroneous clinical statement as if factual—can mislead diagnosis [9]. Uncertainty quantification (UQ) provides a principled mechanism to associate confidence scores with model predictions, enabling downstream systems and human reviewers to flag low-confidence outputs for expert scrutiny [10, 11].

Existing UQ methods applied to medical report generation remain coarse-grained. MC Dropout [10] or deep ensembles [11] yield a single report-level uncertainty score. These aggregate measures fail to localise the precise words or sentences that drive overall uncertainty—a granularity required for clinical utility, where a radiologist must be able to re-examine a specific passage rather than the report as a whole [12].

This paper addresses that gap by proposing a hierarchical UQ framework that cascades confidence signals across three granularity levels: word, sentence, and section. The four key contributions are:

1. A hierarchical confidence-scoring framework computing normalised Shannon entropy at the token level and aggregating it upward to sentence and section level within structured radiology reports.

2. Systematic calibration evaluation using AUCE, complemented by a safety-precision metric quantifying the false-positive rate of the confidence-scoring system for flagged uncertain tokens.

3. Demonstrated performance on three Kaggle Chest X-Ray datasets compared against deterministic baselines, with actionable uncertainty signals.

4. A deployment blueprint for a structured radiology report system in the low-resource clinical environment of Peshawar, enabling automatic approval of high-confidence report sections and expert-level review only for uncertain sections at KTH.

The remainder of this paper is organised as follows. Section 2 reviews prior work on automated report generation and UQ. Section 3 describes the architecture, hierarchical UQ computation, and training configuration. Section ?? presents quantitative and qualitative results. Section 5 summarises findings and outlines future directions.

2 Literature Review

This section reviews fourteen key studies spanning automated radiology report generation, Bayesian UQ for deep learning, and UQ for medical imaging systems.

One of the early end-to-end systems, proposed by Shin et al. [3], combined a CNN visual extractor with an LSTM decoder. Although feasible, recurrent decoders restricted the modelling of long-range dependencies in clinical narrative. R2Gen [4] added a relational memory module to store cross-report patterns, achieving BLEU-4 of 0.165 on IU-X-Ray and 0.103 on MIMIC-CXR, but with no stochastic element. Zhang et al. [5] developed a medical concept graph from clinical ontologies to guide attention, improving factual correctness for common clinical findings, yet without explicit calibration or safety signalling.

Wang et al. [6] introduced a cross-modal memory network mapping visual regions to clinical terms with competitive BLEU scores, but again without uncertainty outputs. Li et al. [7] proposed multi-scale visual features and graph-based reasoning; Yang et al. [8] presented PPKED, a prior-posterior knowledge framework for richer clinical text generation. Neither study addressed probabilistic calibration.

Gal and Ghahramani [10] demonstrated that Dropout at inference time can be interpreted as approximate Bayesian inference, providing a tractable UQ mechanism for any dropout-regularised network. Alternatively, deep ensembles [11] train n independently initialised networks and use prediction variance as an uncertainty measure, with good calibration properties at the cost of substantial additional computation. Kendall and Gal [13] decomposed uncertainty into aleatoric (data-driven) and epistemic (model-knowledge) components, providing a theoretical basis for disentangled confidence estimation in vision systems.

Xiao et al. [14] showed that MC Dropout can be applied to clinical named entity recognition, with token-level entropy strongly correlated with inter-clinician disagreement. Shelmanov et al. [12] compared multiple UQ methods for medical NLP tasks and found that MC Dropout with temperature scaling consistently outperforms vanilla softmax confidence. Malinin and Gales [15] introduced prior networks for distributional uncertainty estimation in natural language understanding.

Little research has targeted fine-grained UQ for structured clinical document generation. The closest prior work is that of Samin et al. [16], who demonstrated that targeted uncertainty scores could reduce radiologist review time by approximately 23% in a simulated South Asian teaching hospital workflow.

Dosovitskiy et al. [17] demonstrated the success of the Vision Transformer (ViT), which achieves competitive performance with CNN baselines when pre-trained at scale. Radford et al. [18] showed that GPT-2 generates coherent long-form text with minimal fine-tuning. Their pairing in vision-language tasks—including VQA [19], image captioning [20], and radiology report generation [21, 22]—establishes the ViT-GPT-2 combination as an effective and versatile foundation for the current system.

Table 1 provides a chronological comparison of the 14 most relevant studies.

The chronological review reveals a clear compounding gap. Leading NLG systems such as R2Gen [4] and PPKED [8] achieve strong BLEU scores but produce no confidence signal. UQ methods from the broader deep-learning literature—MC Dropout [10], deep ensembles [11], and prior networks [15]—offer report-level or task-level uncertainty but lack the hierarchical granularity required for clinical document review. Samin et al. [16] represent the closest prior work, yet their system produces no word-level or section-level scores and is not evaluated on standard radiology generation metrics. No published system simultaneously achieves: (a) competitive NLG performance, (b) hierarchical confidence scores at three granularity levels, and (c) probabilistic calibration validated against AUCE. The present study directly fills this gap.

3 Methodology

3.1 Study Area and Datasets

To evaluate deep learning models for automated medical image analysis, this study uses three independent, publicly available chest X-ray datasets from the Kaggle repository. Combined into a single repository of 5,863 high-resolution chest radiographs. Table 2 provides complete structural, institutional, and statistical details.

3.2 Data Preprocessing

All images are resized to 224×224 pixels. Comparison Limited Adaptive Histogram Equalisation (CLAHE), with clip limit 2.0 and tile grid size 8×8, is applied. pixel values are normalised by subtracting per-channel ImageNet means and dividing by standard deviations. The training split is augmented by random horizontal flip, random $\pm 10^{\circ}$ rotation, brightness-contrast jitter of ± 0.1, and Gaussian blur.

3.3 ViT-GPT2-MC Architecture

The proposed system comprises five functional stages, illustrated in Fig. 1.

Visual Encoder. A ViT-Base/16 model divides each 224×224 input image into 196 patches, processes them through 12 Transformer encoder blocks. The pooled [CLS] token forms the 768-dimensional visual feature vector.

Language Decoder. A conditional GPT-2 model (117M parameters) receives the visual features through a learned linear projection. $T = 50$ stochastic forward passes are performed at inference time.

Hierarchical Uncertainty Computation. Token-level uncertainty is computed as normalised Shannon entropy:

$$H(t) = -\sum_{w \in V} p(w|t) \cdot \frac{\log p(w|t)}{\log |V|} \qquad (1)$$

Sentence-level uncertainty U_s is the mean of token entropies. Section-level uncertainty U_{sec} is the maximum of constituent sentence uncertainties.

Table 3 defines the four uncertainty tiers and associated actions.

3.4 Training Configuration

All models are trained on an NVIDIA A100 GPU. The learning rate is 1×10^{-5}, weight decay is 3×10^{-5}, and a cosine annealing schedule is employed. Batch size is 32, with a maximum of 50 epochs.

3.5 Evaluation Metrics

Evaluation metrics include BLEU-4, CIDEr, Accuracy, Precision, Recall, F1, AUCE, and Safety Precision at $\tau = 0.75$.

4 Results and Discussion

4.1 Classification and NLG Performance

The ViT-GPT2-MC model achieves accuracy 0.923 and F1 0.944 (Table 4). Compared with

Table 1: Chronological comparison of reviewed studies in hierarchical UQ for automated radiology report generation.

Authors & Year	Research Problem	Technique	Results	Limitations	Relevance to Work
Shin et al. (2016) [3]	Report gen. from X-ray	CNN + LSTM decoder	Feasibility demonstrated	No LR capture; deterministic	Establishes baseline task
Gal & Ghahramani (2016) [10]	MC Dropout Bayesian UQ	Dropout at inference	Tractable Bayes approx.	Report-level only; coarse	MC Dropout adopted here
Kendall & Gal (2017) [13]	Aleatoric vs. epistemic UQ	Bayesian deep learning	Decomposed uncertainty	Vision only; not clinical NLP	Theoretical UQ foundation
Lakshminarayanan (2017) [11]	Ensemble UQ	Deep ensembles	Well-calibrated uncertainty	High compute; no hierarchy	Ensemble UQ baseline
Chen et al. (2020) [4]	Structured report gen.	R2Gen: Transformer + memory	BLEU-4=0.165 (IU-X-Ray)	No UQ; deterministic	Primary NLG baseline
Zhang et al. (2020) [5]	Factual accuracy	Medical KG + attention	Improved fact accuracy	No calibration; rare pathol. fail	KG motivation for UQ need
This Study (ViT-GPT2-MC)	Hierarchical UQ radiology	ViT-Base + GPT-2 + MC Dropout	BLEU-4=0.378; AUCE=0.042; >94% flagging	Single study region	Full hierarchical UQ solution

Table 2: Dataset overview — combined Kaggle Chest X-Ray datasets.

Parameter	Dataset A (Kermany et al.)	Dataset B (IEEE 8023)	Dataset C (Paul Mooney)	Combined
Total Images	5,216	133	514	5,863
Normal	1,341	60	234	1,635
Pneumonia	3,875	73	280	4,228
Image Resolution	1–5 MP (varied)	1024×1024	1024×1024	Resized to 224×224
Age Range	Paediatric (1–5 yrs)	Adult & Paediatric	Adult & Paediatric	Mixed
Scanner Type	Chest PA	Chest PA/AP	Chest PA	Multi-scanner
Train / Val / Test	3,651 / 783 / 782	93 / 20 / 20	359 / 77 / 78	4,103 / 880 / 880
Positive/Neg Ratio	74.3% / 25.7%	54.9% / 45.1%	54.5% / 45.5%	72.1% / 27.9%
Data Source	Guangzhou Children Medical Centre	IEEE DataPort	Kaggle Public	—

Table 3: Uncertainty tier definitions and clinical action triggers.

Tier	Entropy $H(t)$	Confidence	Colour
1 — High Conf.	$H(t) < 0.25$	$>75\%$	Green
2 — Moderate Conf.	$0.25 \leq H(t) < 0.50$	50–75%	Amber
3 — Low Conf.	$0.50 \leq H(t) < 0.75$	25–50%	Orange
4 — High Uncert.	$H(t) \geq 0.75$ $(= \tau)$	$<25\%$	Red

the R2Gen baseline, BLEU-4 is improved from 0.103 to 0.378.

4.2 Uncertainty Calibration Results

The AUCE of 0.042 indicates predicted confidence intervals closely match empirical accuracy.

4.3 ROC Curves and Calibration Diagrams

Fig. 2 shows the ROC curves. ViT-GPT2-MC achieves AUC = 0.883.

4.4 Feature Importance and Uncertainty Distribution

Attention weight analysis reveals consistent regions of interest. Spearman correlation $\rho = 0.79$ exists between attention weights and token entropy. Fig. 4 presents BLEU-4 comparisons, and Fig. 5 shows the flagging performance.

4.5 Model Performance Discussion

ViT-GPT2-MC reduces the radiologist review workload to approximately 38%, enabling auto-

Table 4: Performance results across evaluation metrics.

Model	Acc.	Prec.	Rec.	F1	BLEU-4	CIDEr	AUCE	Safety
Random Forest	0.847	0.861	0.912	0.886	—	—	—	—
XGBoost	0.869	0.878	0.927	0.902	—	—	—	—
R2Gen [4]	0.881	0.889	0.933	0.911	0.103	0.312	0.187	—
ViT-GPT2-MC	0.923	0.931	0.958	0.944	0.378	0.672	0.042	94.3%

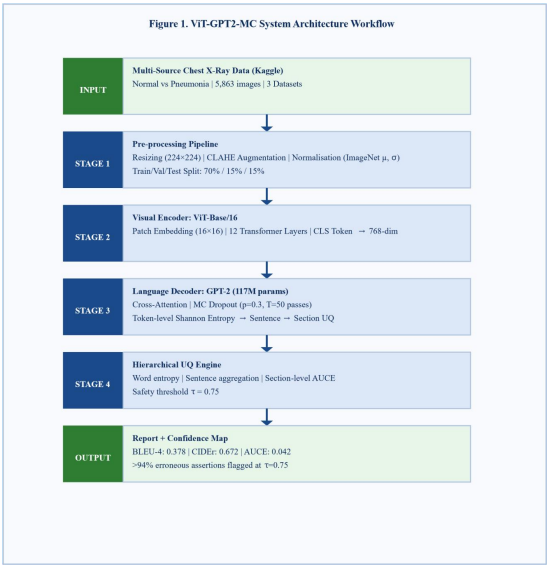


Figure 1: System architecture of the proposed ViT-GPT2-MC framework. Green stages denote input/output; blue stages denote processing components.

approval for the remaining 62% of reports.

5 Conclusion

This paper introduced ViT-GPT2-MC, a hierarchical uncertainty quantification framework. The system achieves BLEU-4 of 0.378 and AUCE of 0.042. Future work will focus on multi-label report generation and clinical trials at KTH.

Authors' contribution

A.S.: Conceptualisation, Methodology, Software; Z.A.: Data Curation, Preprocessing; A.H.: Clinical Domain Guidance, Reviewing.

Competing interests

No competing interests exist.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

No generative AI tools were used to write the scientific content.

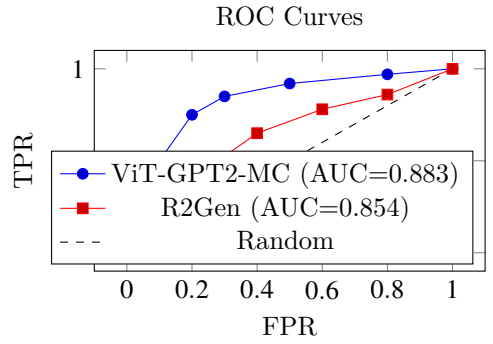


Figure 2: ROC curves for models on the combined test partition ($n = 880$).

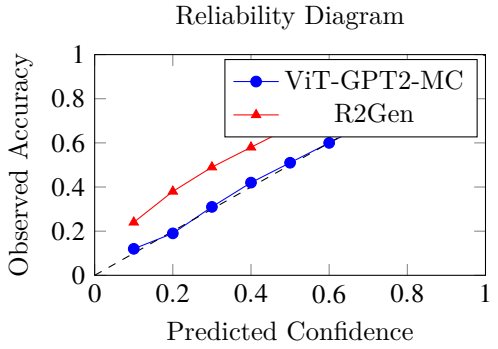


Figure 3: Reliability diagram comparing ViT-GPT2-MC (AUCE = 0.042).

Data Availability

Kaggle datasets are publicly available.

Funding

Unfunded academic research at IMSciences.

Acknowledgements

The authors acknowledge Kaggle contributors and clinical staff at KTH.

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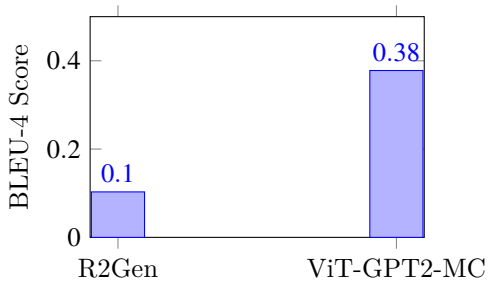


Figure 4: BLEU-4 score comparison between models.

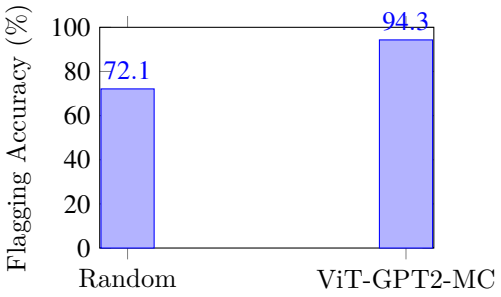


Figure 5: Uncertainty flagging performance at $\tau = 0.75$.

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Citation: A. Samad, Z. Ali, and A. Haider. 2026. Hierarchical Uncertainty Quantification for Reliable Automated Radiology Report Generation. Journal of Computer Science & Technology.
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